

LESSON 3.7

DIVIDING NUMBERS EXPRESSED IN SCIENTIFIC NOTATION



LESSON INTRODUCTION

MA.8.NSO.1.5 Add, subtract, multiply, and divide numbers expressed in scientific notation with procedural fluency.

MA.8.NSO.1.6 Solve real-world problems involving operations with numbers expressed in scientific notation.

LEARNING TARGETS

- I can divide numbers expressed in scientific notation.
- I can solve real-world problems involving operations with numbers expressed in scientific notation.

BENCHMARK COHERENCE

BUILDING ON

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.

WORKING TOWARD

MA.912.NSO.1.1 Extend previous understanding of the laws of exponents to include rational number exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents

ESSENTIAL QUESTIONS

- How do you divide numbers written in scientific notation?
- What range of numbers are possible for the mantissa?
- How does the power of 10 change as you move the decimal in the mantissa to the left/right?

LESSON NARRATIVE

Students discover how to divide numbers expressed in scientific notation. Students are engaged in a collaborative exploration, error analysis activity, guided, and independent practice. Through this repetitive practice students develop the knowledge and skills to divide numbers expressed in scientific notation in mathematical and real-world problems.

COMPONENT SUMMARY

3.7.1 WARM UP (5 MIN.)

Solve and write real-world ratio problems

3.7.2 EXPLORATION (15-20 MIN.)

Make and test conjectures about dividing numbers written in scientific notation

3.7.3 GUIDED INSTRUCTION (10-15 MIN.)

Introduction to how to divide numbers written in scientific notation

3.7.4 GUIDED PRACTICE (10-15 MIN.)

Practice dividing numbers written in scientific notation

3.7.5 WRAP-UP (5 MIN.)

Reflect on mastery of learning targets

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- (Optional) Chart Paper & Markers
- (Optional) Highlighters or Colored Pencils

ADDITIONAL PREPARATION

- Consider creating an anchor chart for dividing numbers written in scientific notation.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not understand why they divide the mantissa and subtract the exponents. This may cause them to divide the exponents and/or subtract the mantissas.
- Students may not understand how to rewrite their answers in scientific notation.

THINGS TO CONSIDER

- Consider reviewing how to convert a fraction to its decimal equivalent by using long division. Remind students that the numerator is the dividend, and the denominator is the divisor.

LITERACY AND LANGUAGE ROUTINES

Collect and Display

In 3.7.2 Exploration students first make conjectures about dividing numbers written in scientific notation. Next, they test, discuss, and revise their conjectures. Debrief the exploration by collecting and displaying a variety of students' revised conjectures. Ask students if they agree or disagree with their peers' conjectures and why.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.3.1 Mathematical Fluency

Students are provided with multiple opportunities to practice dividing numbers written in scientific notation in 3.7.2 Exploration, 3.7.3 Guided Instruction, and 3.7.4 Guided Practice. This repeated practice help students develop the knowledge and skills to accurately and efficiently solve mathematical and real-world problems.

DIFFERENTIATE INSTRUCTION

3.7.2 Exploration provides an auditory entry point as students discuss their conjectures. 3.7.3 Guided Instruction provides a visual entry point as students star the correct answer and circle the error. Provide a visual entry point in 3.7.4 Guided Practice by having students use highlighters or colored pencils to color code the mantissa and powers of 10.

Additional differentiation opportunities are denoted throughout the lesson guide.

$$1. \frac{(5.5 \times 10^2)}{(2.75 \times 10^4)}$$

Mara's work	Colin's work
$\frac{(1.5 \times 10^{11})}{(3 \times 10^9)}$ $= \frac{(1.5)}{(3)} \times \frac{(10^{11})}{(10^9)}$ $= (.5 \times 10^2)$ $= (5 \times 10^2)$	$\frac{(1.5 \times 10^{11})}{(3 \times 10^9)}$ $= 5 \times 10^1$ ★

3.7.1 WARM-UP: TOILET PAPER MATH

DIFFERENTIATE INSTRUCTION

Consider finding prices for packages of toilet paper rolls that match these. Have students determine which they consider as the best buy.

DIFFERENTIATE INSTRUCTION

Consider providing sentence stems to help students structure their explanations. For example:

Since ____ regular rolls are equal to one family mega roll, this number must be estimated to ____, because_____.

QUESTIONING

As time allows, challenge students to swap questions and answer each other's questions. Students can also provide their partner with feedback on their questions. For example, did the question include enough information? Did they find it too easy or too hard?



3.7.1 Warm-Up: Toilet Paper Math

The table shows information from four different toilet paper roll packages.

18 Mega Rolls = 72 Regular Rolls	24 Family Mega Rolls = 123 Regular Rolls
 18 Mega 72 Regular	 24 Family Mega 123 Regular
30 Mega Rolls = 120 Regular Rolls	18 Family Mega Rolls = 90 Regular Rolls
 30 Mega 120 Regular	 18 Family Mega 90 Regular

- How many regular rolls are equal to 1 mega roll?
4 regular rolls
- Explain why determining the number of Regular Rolls packs that are equal to 1 Family Mega Rolls pack would require you to estimate.
Answers will vary. In one of the packages there are 5.125 regular rolls per family mega roll but in the other package there are 5 regular rolls per family mega roll. Because the ratios are not equivalent, the number of regular rolls equal to 1 family mega roll requires estimation. For example, there are about 5 or 5.1 regular rolls in a family mega roll.
- Write a question about the toilet paper rolls that could be asked to a 6th grader.
Answers will vary. Which size gets you the most toilet paper for your money? (best unit price)



3.7.2 Exploration: Making a Conjecture for How to Divide Numbers Expressed in Scientific Notation

Work with your partner to complete the problem in each table. First, determine the answer to the division problem using whole numbers in the first row. Then, convert the whole numbers to scientific notation in the second row.

1.	Problem	Answer
Whole Numbers	$600 \div 25$	24
Scientific Notation	$(6.0 \times 10^2) \div (2.5 \times 10^1)$	2.4×10^1

2.	Problem	Answer
Whole Numbers	$4500 \div 30$	150
Scientific Notation	$(4.5 \times 10^3) \div (3.0 \times 10^1)$	1.5×10^2

3.	Problem	Answer
Whole Numbers	$7,000,000 \div 28,000$	250
Scientific Notation	$(7.0 \times 10^6) \div (2.8 \times 10^4)$	2.5×10^2

4.	Problem	Answer
Whole Numbers	$70,000 \div 28,000,000$	0.0025
Scientific Notation	$(7.0 \times 10^4) \div (2.8 \times 10^7)$	2.5×10^{-3}

5.	Problem	Answer
Whole Numbers	$90,000 \div 180,000$	0.5
Scientific Notation	$(9.0 \times 10^4) \div (1.8 \times 10^5)$	5×10^{-1}

- What do you notice about the problems using scientific notation and their corresponding answers?
Answers will vary, but could include how you divide normally and then subtract the exponents for the tens.
- Make a conjecture about how to divide numbers written in scientific notation.
Group the values together and divide, group the tens together and subtract their exponents.
- Ask two other pairs of partners for the conjecture they wrote. Listen to their conjecture and write a summary. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating that your summary was accurate.

My summary of the conjecture	Initials
<i>Answers will vary.</i>	
<i>Answers will vary.</i>	

- Revisit your conjecture with your partner and resolve any differences if necessary.

3.7.2 EXPLORATION: MAKING A CONJECTURE FOR HOW TO DIVIDE NUMBERS EXPRESSED IN SCIENTIFIC NOTATION

TEACHER MOVES

Notice and Wonder

Ask students what they notice and wonder about the relationships between the mantissa of the dividend, divisor, and quotient. Also, prompt students to share what they notice and wonder about the power of 10 of the dividend, divisor, and quotient.

ENRICHMENT

How did your original conjecture change after talking to your classmates? What prompted the changes you made in your conjecture? How can you be sure that your conjecture is accurate?

TEACHER MOVES

Collect and Display

Students first make conjectures about dividing numbers written in scientific notation. Next, they test, discuss, and revise their conjectures. Debrief the exploration by collecting and displaying a variety of students' revised conjectures. Ask students if they agree or disagree with their peers' conjectures and why.

3.7.3 GUIDED INSTRUCTION: DIVIDING NUMBERS WRITTEN IN SCIENTIFIC NOTATION

TEACHER MOVES

This activity scaffolds the learning for students. Consider a back-and-forth of reading the question and having students think through the answer and engage in a discussion for each sub-question.

ENRICHMENT

When is a quotient negative? Is the dividend or divisor negative? What does a negative power of 10 imply about the standard form of the number? When is a quotient greater/less than one?

DIFFERENTIATE INSTRUCTION

To help students identify and correct the error, consider prompting students to use a checklist to review Mara's work. For example: Process Mara's Work. For example:

!!!

Write the problem in scientific notation Separate the decimals and the powers of 10 Divide mantissas Subtract powers of 10 Convert final answer to scientific notation Starring the correct answer and circling the error provides a visual entry point.

TEACHER MOVES

Consider providing sentence stems to help students structure their explanations. For example:

The mistake you made was _____. To avoid making this mistake, I recommend that you _____.



3.7.3 Guided Instruction: Dividing Numbers Written in Scientific Notation

1. Gizmo Gadgets produced 50,000 widgets last month. They shipped them out to 2,500 different stores.
 - a. How many widgets did each store receive? Write your answer in scientific notation.

$$(5.0 \times 10^4) \div (2.5 \times 10^3) = 2.0 \times 10^1$$
 - b. Determine if the following equation is true. Support your answer using mathematics.

$$2 \times 10^1 = \frac{(5 \times 10^4)}{(2.5 \times 10^3)}$$

Yes, it's true because $\frac{5}{2.5} = 2$ and $\frac{10^4}{10^3} = 10^1$
 - c. Explore what happens when the dividend is less than the divisor by completing the problem.

$$\frac{(7 \times 10^4)}{(2.8 \times 10^6)} = 2.5 \times 10^{4-6} = 2.5 \times 10^{-2}$$
 - d. Discuss with your partner why the answer to Part C is not a negative number and has a negative exponent.
 - e. Solve by filling in the blanks.

$$\frac{(-7 \times 10^4)}{(2.8 \times 10^6)} = -2.5 \times 10^{4-6} = -2.5 \times 10^{-2}$$
 - f. Discuss with your partner why the answer to Part E is a negative number.

2. Mara and Colin were given the following problem. Their work is shown.

The distance from the sun to the Earth is about 150,000,000,000 meters. The speed of light is (3×10^9) meters per second. How many seconds does it take for light from the sun to reach the Earth?

Mara's work	Colin's work
$\frac{(1.5 \times 10^{11})}{(3 \times 10^9)}$ $= \frac{(1.5)}{(3)} \times \frac{(10^{11})}{(10^9)}$ $= (.5 \times 10^2)$ $= (5 \times 10^2)$	$\frac{(1.5 \times 10^{11})}{(3 \times 10^9)}$ $= 5 \times 10^1$



- a. Place your initials by the student's name who has the correct answer in the table.
- b. Circle and correct the error made by the student with the incorrect answer.

$$0.5 \times 10^2 = 5.0 \times 10^1$$

Write a note to the person who is incorrect explaining their mistake.

Means

I

Start

To

Acquire

Knowledge

Experience

Skills

Answers will vary but could include: maybe writing 0.5 would help the decimal point to stand out so you know you have to move it and change the exponent of the 10 to put in proper scientific notation.



3.7.4 Guided Practice: Finding Quotients

Find the quotient for each.

1. $\frac{(5.5 \times 10^2)}{(2.75 \times 10^4)}$

2.0×10^{-2}

2. $\frac{(7.2 \times 10^{26})}{(8 \times 10^{26})}$

$0.9 \times 10^0 = 9.0 \times 10^{-1}$

3. $\frac{(2.16 \times 10^{-1})}{(2 \times 10^{-2})}$

1.08×10^1

4. $\frac{(9.87 \times 10^{-4})}{(4.7 \times 10^{-2})}$

2.1×10^{-2}

3.7.4 GUIDED PRACTICE: FINDING QUOTIENTS

TEACHER MOVES

Students are provided with multiple opportunities to practice dividing numbers written in scientific notation in 3.7.2 Exploration, 3.7.3 Guided Instruction, and 3.7.4 Guided Practice. This repeated practice helps students develop the knowledge and skills to accurately and efficiently solve mathematical and real-world problems.

DIFFERENTIATE INSTRUCTION

Provide a visual entry point by having students use highlighters or colored pencils to color code the mantissa and powers of 10.



3.7.5 Wrap-Up: Reflection

1. Choose the emoji that reflects your understanding of dividing numbers expressed in scientific notation.



0 1 2 3 4 5 6 7 8 9 10

2. Complete the statement.

I chose this emoji because _____

 _____.

Answers will vary.

3.7.5 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

Consider providing options for how students respond to question 2. For example, they could write and solve a problem, provide a written explanation, or record their justification with an audio recorder (i.e., Flipgrid).